draft 10 final draft with fixed visualizations

uploading and cleaning data

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.3     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(lubridate)
library(kableExtra)
```

```
Warning: package 'kableExtra' was built under R version 4.3.3


Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

    group_rows
```

```r
# cleaning function
clean_pm25 <- function(file, city_name, poc_value) {
  read_csv(file) |>
    filter(POC == poc_value) |>
    select(Date, `Daily Mean PM2.5 Concentration`) |>
    mutate(Date = as.Date(Date, format = "%m/%d/%Y")) |>
    rename(!!paste0(city_name, " PM2.5") := `Daily Mean PM2.5 Concentration`)
}
# load all datasets
PM25_data <- list(
  Boston = list(
    `2018` = clean_pm25("PM25_Boston_2018.csv", "Boston", 1),
    `2019` = clean_pm25("PM25_Boston_2019.csv", "Boston", 1),
    `2020` = clean_pm25("PM25_Boston_2020.csv", "Boston", 1),
    `2021` = clean_pm25("PM25_Boston_2021.csv", "Boston", 1),
    `2022` = clean_pm25("PM25_Boston_2022.csv", "Boston", 3)
  ),
  Worcester = list(
    `2018` = clean_pm25("PM25_Worcester_2018.csv", "Worcester", 3),
    `2019` = clean_pm25("PM25_Worcester_2019.csv", "Worcester", 3),
    `2020` = clean_pm25("PM25_Worcester_2020.csv", "Worcester", 3),
    `2021` = clean_pm25("PM25_Worcester_2021.csv", "Worcester", 3),
    `2022` = clean_pm25("PM25_Worcester_2022.csv", "Worcester", 3)
  ),
  Brockton = list(
    `2018` = clean_pm25("PM25_Brockton_2018.csv", "Brockton", 3),
    `2019` = clean_pm25("PM25_Brockton_2019.csv", "Brockton", 3),
    `2020` = clean_pm25("PM25_Brockton_2020.csv", "Brockton", 3),
    `2021` = clean_pm25("PM25_Brockton_2021.csv", "Brockton", 3),
    `2022` = clean_pm25("PM25_Brockton_2022.csv", "Brockton", 3)
  ),
  NorthAdams = list(
    `2018` = clean_pm25("PM25_NorthAdams_2018.csv", "North Adams", 3),
    `2019` = clean_pm25("PM25_NorthAdams_2019.csv", "North Adams", 3),
    `2020` = clean_pm25("PM25_NorthAdams_2020.csv", "North Adams", 3),
    `2021` = clean_pm25("PM25_NorthAdams_2021.csv", "North Adams", 3),
    `2022` = clean_pm25("PM25_NorthAdams_2022.csv", "North Adams", 3)
  )
)
```

```
Rows: 95 Columns: 22
-- Column specification --------------------------------------------------------
```

```
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 114 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 454 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 827 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 816 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 446 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 417 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 413 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 354 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 346 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 416 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

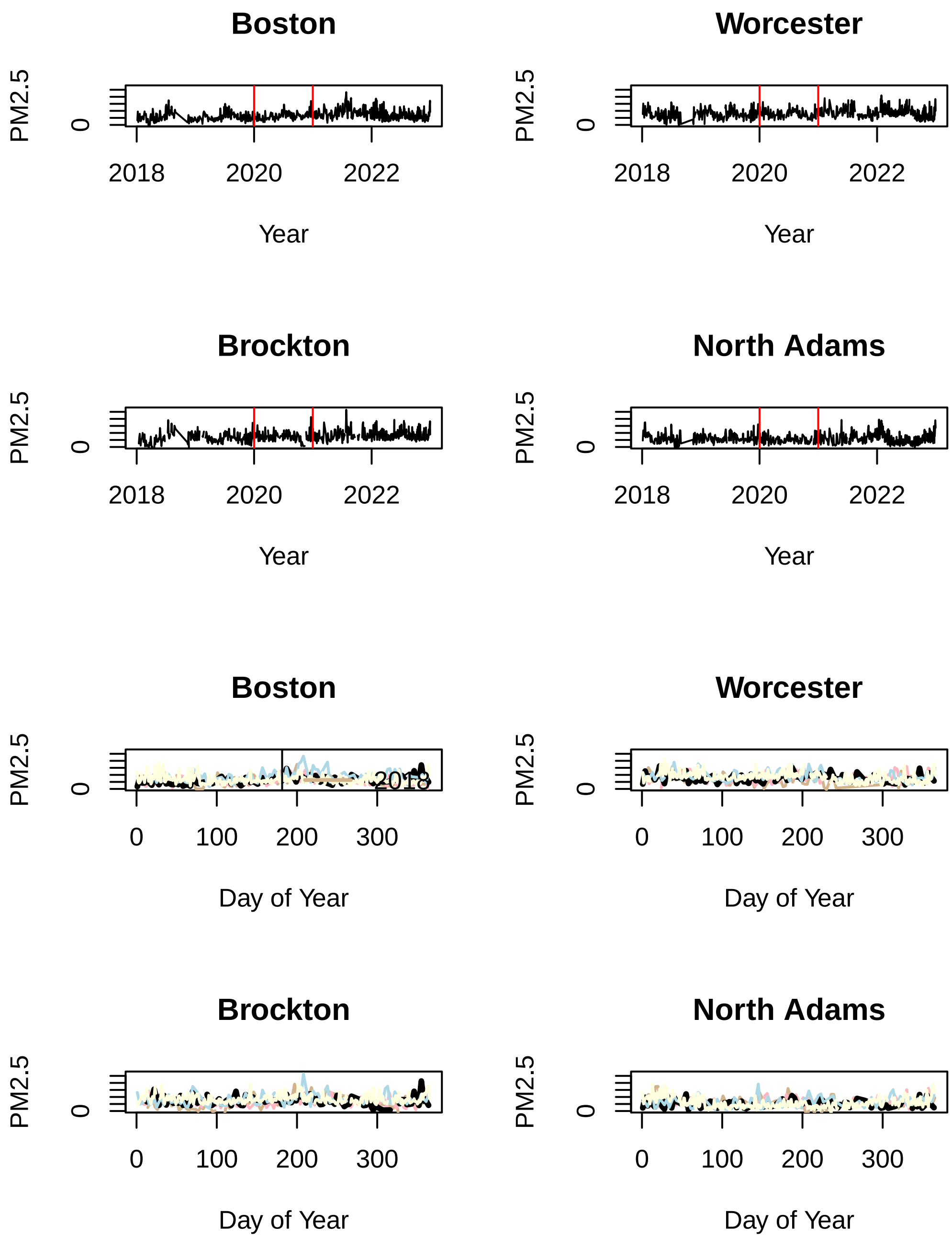
```
Rows: 407 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 406 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 329 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 349 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 351 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 348 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
```

```
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 350 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 360 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Rows: 351 Columns: 22
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# function to build a full year
build_year <- function(year, data_list) {

  # full calendar
  dates <- tibble(
    Date = seq(as.Date(paste0(year, "-01-01")),
               as.Date(paste0(year, "-12-31")),
               by = "day")
  )

  # extract that year from each city
  dfs <- map(data_list, ~ .x[[as.character(year)]])
```

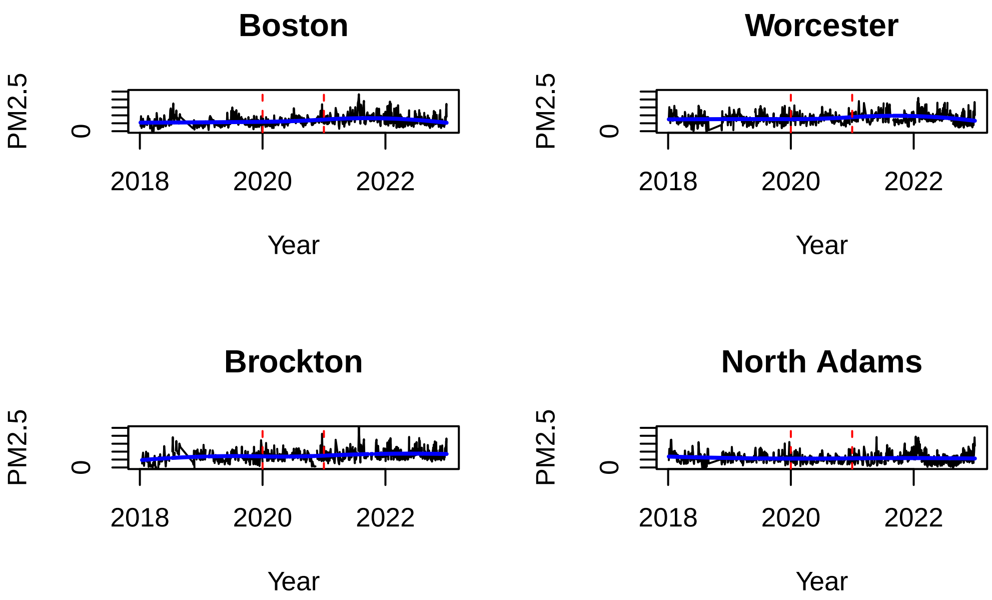
```r
  # join all cities onto full calendar
  out <- reduce(dfs, left_join, by = "Date")

  # attach year column
  out <- out |> mutate(Year = year)

  return(out)
}

# full dataset of all years
years <- 2018:2022

all_PM25 <- map_dfr(years, ~ build_year(.x, PM25_data))

# time index
all_PM25 <- all_PM25 |>
  arrange(Date) |>
  mutate(Date_Number = as.integer(Date - min(Date)))
# convert to 3 day spacing to smooth
all_PM25_3day <- all_PM25 |>
  filter(Date_Number %% 3 == 0)
```

plots, visuals



Regression splines

with splines set for jan 1 - dec 31 2020 so that we can see how the seasonal trends of air pollution were impacted



summary tables of regression spline fits

make pretty with figures and kable()

```r
print(summary(fit_boston))
```

```
Call:
lm(formula = x ~ t + t2_k1 + t2_k2, na.action = na.exclude)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.0657 -2.2209 -0.5828  1.5726 16.4733 

Coefficients: (2 not defined because of singularities)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 6.6832282  0.1118644   59.74  < 2e-16 ***
t           0.0009741  0.0002034    4.79 1.99e-06 ***
t2_k1              NA         NA      NA       NA
t2_k2              NA         NA      NA       NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.15 on 791 degrees of freedom
Multiple R-squared:  0.02819,   Adjusted R-squared:  0.02696
F-statistic: 22.95 on 1 and 791 DF,  p-value: 1.989e-06
```

```r
print(summary(fit_worcester))
```

```
Call:
lm(formula = x ~ t + t2_k1 + t2_k2, na.action = na.exclude)

Residuals:
   Min     1Q Median     3Q    Max
-8.269 -2.370 -0.399  1.914 12.578

Coefficients: (2 not defined because of singularities)
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 8.2494269  0.1243772  66.326   <2e-16 ***
t           0.0005189  0.0002245   2.312   0.0211 *
t2_k1              NA         NA      NA       NA
t2_k2              NA         NA      NA       NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.426 on 757 degrees of freedom
Multiple R-squared:  0.007009,  Adjusted R-squared:  0.005697
F-statistic: 5.343 on 1 and 757 DF,  p-value: 0.02107
```

```r
print(summary(fit_brockton))
```

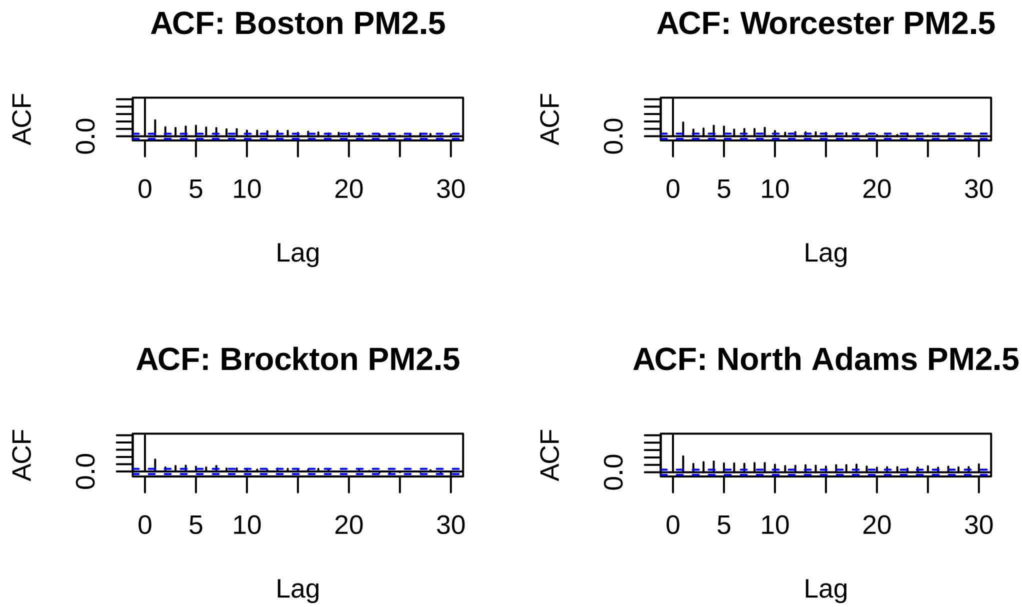
```
Call:
lm(formula = x ~ t + t2_k1 + t2_k2, na.action = na.exclude)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-7.3095 -2.2540 -0.5633  1.8369 18.4284 

Coefficients: (2 not defined because of singularities)
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 7.8047902  0.1209381  64.535  < 2e-16 ***
t           0.0018107  0.0002227   8.132 1.77e-15 ***
t2_k1              NA         NA      NA       NA    
t2_k2              NA         NA      NA       NA    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.298 on 742 degrees of freedom
Multiple R-squared:  0.08184,   Adjusted R-squared:  0.0806 
F-statistic: 66.13 on 1 and 742 DF,  p-value: 1.767e-15
```

```r
print(summary(fit_northadams))
```

```
Call:
lm(formula = x ~ t + t2_k1 + t2_k2, na.action = na.exclude)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.0977  -2.4018  -0.6237   1.7748  13.6910 

Coefficients: (2 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  5.8812503  0.1215996  48.366   <2e-16 ***
t           -0.0002277  0.0002202  -1.034    0.301    
t2_k1               NA         NA      NA       NA    
t2_k2               NA         NA      NA       NA    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.372 on 767 degrees of freedom
Multiple R-squared:  0.001392,  Adjusted R-squared:  9.052e-05 
F-statistic:  1.07 on 1 and 767 DF,  p-value: 0.3014
```

WHITE NOISE CHECK



make a table and reference the figure

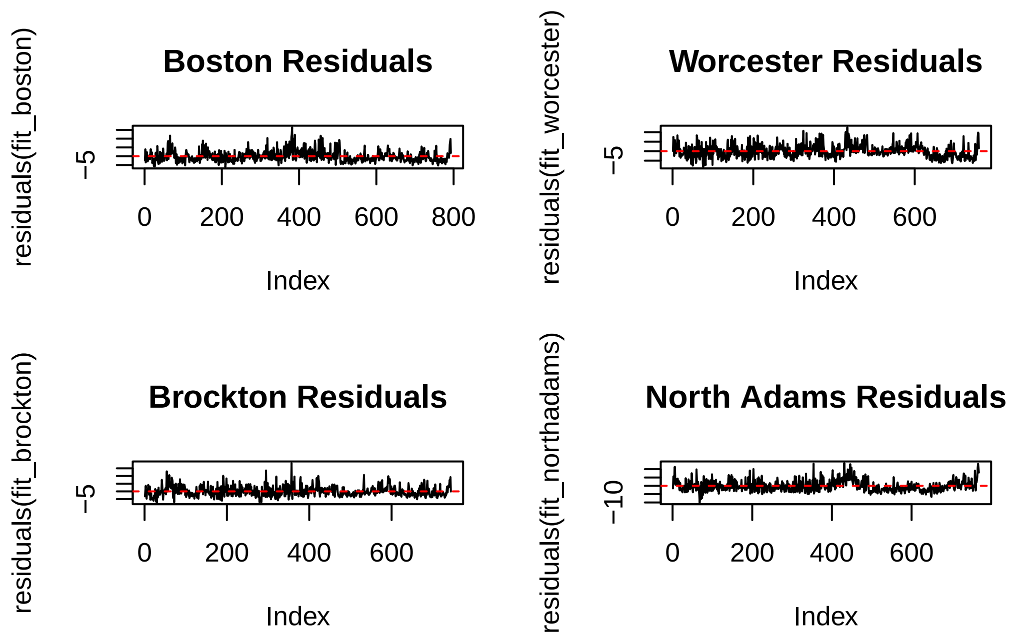
Ljung-Box Test

```
    Box-Ljung test

data:  res_boston
X-squared = 616.42, df = 16, p-value < 2.2e-16


    Box-Ljung test

data:  res_worcester
X-squared = 469.22, df = 16, p-value < 2.2e-16


    Box-Ljung test

data:  res_brockton
X-squared = 197.87, df = 16, p-value < 2.2e-16
```

```
    Box-Ljung test

data:  res_northadams
X-squared = 720.06, df = 16, p-value < 2.2e-16
```

Table 1: This table shows the results of conducting the Box-Ljung Test with PM2.5 data from Boston, Worcester, Brockton, and North Adams.

City	X.squared	df	p.value
Boston	186.190	16	< 2.2e-16
Worcester	84.003	16	= 3.135e-11
Brockton	74.384	16	= 1.682e-09
North Adams	116.330	16	< 2.2e-16



ADF Test

Regular Linear Regression

```
    Augmented Dickey-Fuller Test
```

```
data:  residuals(fit_boston)
Dickey-Fuller = -5.2543, Lag order = 9, p-value = 0.01
alternative hypothesis: stationary


    Augmented Dickey-Fuller Test

data:  residuals(fit_worcester)
Dickey-Fuller = -4.9585, Lag order = 9, p-value = 0.01
alternative hypothesis: stationary


    Augmented Dickey-Fuller Test

data:  residuals(fit_brockton)
Dickey-Fuller = -5.9523, Lag order = 9, p-value = 0.01
alternative hypothesis: stationary


    Augmented Dickey-Fuller Test

data:  residuals(fit_northadams)
Dickey-Fuller = -3.9999, Lag order = 9, p-value = 0.01
alternative hypothesis: stationary
```